

CSC1110 Assignment

October, 2025

Description

Modern chatbots are based on Language Language Models (LLMs) and are being increasingly adopted within organisations to assist both their employees and their target user-base. Suppose you have been asked by your organisation to assess the feasibility of integrating a chatbot into their official website. This chatbot would be used to assist your target user-base with queries they may have.

Now suppose that the organisation that wants to deploy this system is the Irish Election Commission. The Commission wants to use one of these models to provide users with unbiased, factual and informative answers to questions about the 2025 Irish presidential elections in an automated fashion.

The system must adhere to election laws. Specifically, regulations around election information are described in Part 5 of the Election Reforms Act 2022¹. Systems that are used to provide information to the electorate must adhere to these regulations.

Given the above requirements, choose any commercial or open-source chatbot you like (e.g. ChatGPT, Gemini etc.) and perform a feasibility study which assesses the suitability of this system to the above use case. This should be done through systematic testing and should include a variety of test prompts along with an analysis responses. The final report should include the following sections:

1. **Overview of Assessment:** Provide an overview of your assessment.
(5 marks)
2. **Approach to testing:** Describe the structure of your assessment, including the range of tests performed and the motivation for this particular set of tests. Screenshots of all prompts and responses should be included in the **Appendix** section of your report, grouped by the aspect of the system

¹<https://www.irishstatutebook.ie/eli/2022/act/30/enacted/en/print#part5>

you wanted to test.
(30 marks)

3. **Data Insights:** Describe what the tests you have performed reveal about the data used to train this model and how the outputs of these tests reveal this. Describe how you think the training data has affected the model based on what you have learned about approaches to language modeling. Note anything important you found that is relevant to this use case.
(40 marks)
4. **Overall Assessment:** With reference to the above results, discuss whether you think this system would be appropriate to use for this application and why/why not.
(25 marks)

Deadline and Submission Details

The deadline for this assignment is **Friday the 26th of November @ 23:59**. The template for the project is available on the Loop assignment submission webpage. This includes word counts for each section. Please include your name and student number within file, as shown in template.

Please submit your report as a single PDF file with the following file name structure:

`<surname>_<student number>_<module code>.pdf`
(e.g. `rushe_12345678_csc1122.pdf`)

Marking Rubric

Below is a description of the level of achievement expected for each grading band.

- **Excellent** ($\geq 70\%$): An extremely well-structured and systematic approach to testing with a deep level of insight. Demonstrates sophisticated understanding of language models and the factors that affect their output. Comprehensive references to relevant external literature used to inform analysis. This level of analysis would be of interest and insightful to the broader academic community if published.
- **Very Good** ($\geq 60\%$): A well-structured and systematic approach to testing with a very good level of insight. Demonstrates generally very good understanding of language models and the factors that affect their output. Some references to relevant external literature used to inform

analysis. This level of analysis would be of interest and insightful to the broader academic community given a slightly more detailed analysis or more grounding in existing literature.

- **Good** ($\geq 50\%$) A generally well-structured approach to testing with a good level of insight, through perhaps missing some key considerations in test design. Demonstrates generally good understanding of language models and the factors that affect their output, though might be lacking understanding in some places. A limited number of references to relevant external literature used to inform analysis. This analysis is generally good but would require more work to determine whether the system is truly appropriate for the given use-case.
- **Satisfactory** ($\geq 40\%$): An ok approach to testing with surface levels of insight, through missing a considerable number of important considerations in test design. Generally demonstrates basic understanding of language models and the factors that affect them, though substantially lacking understanding in some places. Little to no relevant external literature used to inform analysis. This analysis overall is basic and would require a considerable amount of additional work to determine whether the system is truly appropriate for the given use-case.
- **Unsatisfactory** ($< 40\%$): An insufficient approach to testing with overly vague and inconclusive insights. Missing most of the important considerations for test design. Demonstrates little/incorrect understanding of language models and the factors that affect them. Little to no relevant external literature used to inform analysis. This analysis is completely insufficient and would not be appropriate to gain insights on the use of chatbots for this application.